

PYTHON - SE - BACKEND

SOFTWARE ENGINEERING – MODULE 4

Database Management – SQL & PL/SQL

SESSION / ASSIGNMENT – 4

TASKS :

1. **Create a table called Playlist with columns: id, song_name, artist, genre, and play_count. Insert at least 5 sample records representing your favorite songs from Spotify.**

→ CODE:

```
CREATE TABLE Playlist (  
    id INT PRIMARY KEY AUTO_INCREMENT,  
    song_name VARCHAR(100),  
    artist VARCHAR(100),  
    genre VARCHAR(50),  
    play_count INT  
);  
INSERT INTO Playlist (song_name, artist, genre, play_count)  
VALUES  
('Blinding Lights', 'The Weeknd', 'Pop', 250),  
('Shape of You', 'Ed Sheeran', 'Pop', 320),  
('Believer', 'Imagine Dragons', 'Rock', 180),  
('Perfect', 'Ed Sheeran', 'Romantic', 275),  
('Heat Waves', 'Glass Animals', 'Alternative', 210);  
SELECT * FROM Playlist;
```

OUTPUT:

Server: MySQL-3306 » Database: instadone » Table: Playlist

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0118 seconds.)

```
CREATE TABLE Playlist ( id INT PRIMARY KEY AUTO_INCREMENT, song_name VARCHAR(100), artist VARCHAR(100), genre VARCHAR(50), play_count INT );
```

5 rows inserted.
Inserted row id: 5 (Query took 0.0022 seconds.)

```
INSERT INTO Playlist (song_name, artist, genre, play_count) VALUES ('Blinding Lights', 'The Weeknd', 'Pop', 250), ('Shape of You', 'Ed Sheeran', 'Pop', 320), ('Perfect', 'Ed Sheeran', 'Romantic', 275), ('Heat Waves', 'Glass Animals', 'Alternative', 210);
```

Showing rows 0 - 4 (5 total, Query took 0.0003 seconds.)

```
SELECT * FROM Playlist;
```

Number of rows: 25 Filter rows: Search this table Sort by key: None

	id	song_name	artist	genre	play_count
☐ Edit ☐ Copy ☐ Delete	1	Blinding Lights	The Weeknd	Pop	250
☐ Edit ☐ Copy ☐ Delete	2	Shape of You	Ed Sheeran	Pop	320
☐ Edit ☐ Copy ☐ Delete	3	Believer	Imagine Dragons	Rock	180
☐ Edit ☐ Copy ☐ Delete	4	Perfect	Ed Sheeran	Romantic	275
☐ Edit ☐ Copy ☐ Delete	5	Heat Waves	Glass Animals	Alternative	210

2. Write a SELECT query to display only the song_name and artist from the Playlist table, but show the artist column as 'Singer' using an alias.

→ CODE:

```
SELECT song_name, artist AS Singer  
FROM Playlist;
```

OUTPUT:

Showing rows 0 - 4 (5 total, Query took 0.0023 seconds.)

```
SELECT song_name, artist AS Singer FROM Playlist;
```

Number of rows: 25 Filter rows: Search this table

	song_name	Singer
☐ Edit ☐ Copy ☐ Delete	Blinding Lights	The Weeknd
☐ Edit ☐ Copy ☐ Delete	Shape of You	Ed Sheeran
☐ Edit ☐ Copy ☐ Delete	Believer	Imagine Dragons
☐ Edit ☐ Copy ☐ Delete	Perfect	Ed Sheeran
☐ Edit ☐ Copy ☐ Delete	Heat Waves	Glass Animals

3. Write a **SELECT** query to find all songs in the Playlist table where the genre is 'Pop' and play_count is greater than 100. Sort the results by play_count in descending order.

➔ **CODE:**

```
SELECT song_name, artist, genre, play_count
FROM Playlist
WHERE genre = 'Pop'
AND play_count > 100
ORDER BY play_count DESC;
```

OUTPUT:

Show query box

✓ Showing rows 0 - 1 (2 total, Query took 0.0012 seconds.) [play_count: 320... - 250...]

```
SELECT song_name, artist, genre, play_count FROM Playlist WHERE genre = 'Pop' AND play_count > 100 OR
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	song_name	artist	genre	play_count
☐	Shape of You	Ed Sheeran	Pop	320
☐	Blinding Lights	The Weeknd	Pop	250

4. Use the **COUNT** aggregate function to find out how many songs in your Playlist table belong to the genre 'Hip-Hop'.

➔ **CODE:**

```
SELECT COUNT(*) AS hip_hop_songs
FROM Playlist
WHERE genre = 'Hip-Hop';
```

OUTPUT:

Your SQL query has been executed successfully.

```
SELECT COUNT(*) AS hip_hop_songs FROM Playlist WHERE genre = 'Hip-Hop';
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Extra options

hip_hop_songs
0

5. Write a SELECT query that shows the total number of plays (SUM of play_count) for each genre in your Playlist table, grouped by genre.

Hint: Use GROUP BY with the SUM function to aggregate play counts by genre.

➔ CODE:

```
SELECT genre, SUM(play_count) AS total_plays
FROM Playlist
GROUP BY genre;
```

OUTPUT:

✓ Showing rows 0 - 3 (4 total, Query took 0.0009 seconds.)

```
SELECT genre, SUM(play_count) AS total_plays FROM Playlist GROUP BY genre;
```

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

☐ Show all | Number of rows: 25 ▼ Filter rows:

Extra options

			genre	total_plays
☐	Edit	Copy	Delete Pop	570
☐	Edit	Copy	Delete Rock	180
☐	Edit	Copy	Delete Romantic	275
☐	Edit	Copy	Delete Alternative	210